

Designing a Robust Social Media-Driven Index for Measuring Customer Satisfaction Through Weighted Emotions and Behavioral Silences

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Abstract—Traditional customer satisfaction metrics rely heavily on structured surveys and therefore struggle to capture the dynamic, unstructured, and spontaneous nature of customer feedback on social media. This paper proposes a novel Customer Satisfaction Index (CSI) specifically designed for social media interactions, enabling continuous and automated monitoring of individual customer satisfaction. The proposed CSI integrates sentiment intensity derived from labeled messages, applies exponential temporal decay to emphasize recent interactions, and incorporates explicit penalty terms for emotionally neutral messages and prolonged periods of communication silence, which may indicate latent dissatisfaction. The index is formulated as a time-weighted average of emotional scores, normalized by the sum of temporal weights augmented with neutrality and silence penalties. A pilot validation study was conducted using real-world customer service chat logs. From 100 eligible customers, 36 complete CSI–survey pairs were obtained. On this pilot sample, the CSI demonstrates strong alignment with survey-based Overall Satisfaction, achieving a Pearson correlation of $r = 0.817$, $R^2 = 0.667$. Although preliminary, these results suggest that the proposed CSI is a promising real-time proxy for survey-based customer satisfaction and supports its potential for early-warning and proactive intervention in social customer service settings.

Keywords—Customer Satisfaction Index, Social Network Behavior, Sentiment Analysis.

I. INTRODUCTION

Organizations traditionally rely on survey-based metrics like CSAT, NPS, and ACSI to monitor service quality and loyalty [1]. While structured and interpretable, these instruments are inherently solicited, episodic, and lagging; responses are often sparse, biased toward extremes, and fail to capture the silent majority [2], [3]. As interactions shift toward digital chat and social messaging, survey-only approaches increasingly prove insufficient to capture the fine-grained, temporal dynamics of customer happiness in real-world service journeys [4], [5]. Advances in NLP enable large-scale sentiment analysis, with indices like NSS and ABSA providing actionable diagnostics [4], [5]. However, two gaps persist: (i) most

indices are *static aggregates* that ignore within-conversation trajectory and recency [5], [6]; and (ii) they fail to capture *silent dissatisfaction*—long gaps or repetitive transactional messages signaling disengagement [7]. Consequently, CX practice is shifting toward *early-warning systems* that require real-time, conversation-level satisfaction scores [8], [9]. Such systems must be sensitive to both explicit sentiment shifts and implicit disengagement as messages arrive. However, naively recomputing time-decayed scores in high-volume environments is computationally prohibitive, necessitating more efficient, online-updated measures. We propose a *time-weighted Customer Satisfaction Index (CSI)* for service conversations that (i) applies exponential decay for recency prioritization, (ii) integrates utterance-level sentiment intensity, and (iii) penalizes neutral messages and communication gaps to capture “silent dissatisfaction”. Employing a recursive exponential aggregation scheme enables $O(1)$ complexity, generalizing NSS-style indices by embedding disengagement into a single score. We validate CSI on chat logs from 100 customers (36 survey pairs) against CSAT, NPS, and Happiness using correlation, regression, ROC/AUC, and stratified evaluations.

The remainder of this paper is organized as follows. Section II reviews literature on survey-based satisfaction measurement, sentiment and happiness indices, aspect-based sentiment analysis, and hybrid VoC frameworks. Section III details the proposed CSI formulation, exponential aggregation scheme, and survey design. Section IV presents our empirical setup and results, including ablation and stratified analyses. Section V discusses implications and limitations, and Section VI concludes with directions for future research and real-world applications.

II. LITERATURE REVIEW

Classical customer satisfaction research has largely been grounded in survey-based instruments such as CSAT, NPS, and the American Customer Satisfaction Index (ACSI) [1]. These models conceptualize satisfaction through latent constructs (e.g.,

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expectations, perceived quality, perceived value) and estimate them via structured questionnaires. While well-established and extensively validated, such instruments are inherently solicited and episodic: they capture customer evaluations at discrete points in time and are susceptible to response bias and temporal lag [2], [3]. A substantial proportion of customers either ignore survey invitations or respond only when experiences are extremely positive or negative, leaving a blind spot in the middle of the satisfaction spectrum.

To address these limitations, recent work integrates survey data with signals from social media, review platforms, and CRM logs in hybrid Voice-of-Customer (VoC) frameworks [4], [5], [8]. Net Sentiment Score (NSS) has been proposed as a sentiment-based analogue and leading indicator of NPS. It is computed from the balance of positive versus negative mentions [10], [11]. However, empirical studies report only moderate correlations, suggesting that while unsolicited sentiment can anticipate loyalty shifts, it does not fully replace traditional surveys [12]. Parallel research on Social CSI/SCSI extends indices such as ACSI by incorporating social-media sentiment and other unstructured signals [8], [13], typically aggregating at the brand or product level and relating scores to repurchase, churn, or revenue, sometimes via Kano-style analyses of must-be, performance, and excitement attributes [13]. However, these approaches still operate at coarse temporal and interactional resolutions and do not explicitly capture within-conversation dynamics or the silent-dissatisfaction patterns characteristic of real-time customer service [5], [14]. Advances in NLP, from lexicons to transformers (e.g., BERT), enable extracting satisfaction signals directly from unstructured text [5], [6]. While sentiment pipelines and Aspect-Based Sentiment Analysis (ABSA) provide granular diagnostics [15], existing indices remain largely *static*. They often aggregate mentions within windows rather than modeling conversations as ordered sequences where recency and trajectory are paramount [14]. Furthermore, few metrics explicitly encode silent dissatisfaction—such as long communication gaps or repetitive transactional chatter—as negative signals [16].

This work extends the literature by formulating a conversation-level, time-weighted CSI that incorporates (i) sentiment intensity, (ii) normalized indexing (akin to NSS), and (iii) explicit penalties for temporal decay and silent dissatisfaction [14], [16]. Unlike brand-level aggregates, our model is tailored for real-time monitoring of individual customer journeys, directly linking unsolicited conversational signals with established satisfaction constructs for empirical validation.

III. METHODOLOGY

A. Time-Weighted CSI Formulation

We formalize the proposed Customer Satisfaction Index (CSI) at the level of an individual conversation between a customer and the service. Let the conversation contain n_e messages with non-neutral emotional content. For each such

message i , we denote by $r_{e_i} \in [-1, 1]$ its emotional intensity and by w_{t_i} its temporal weight. The CSI is defined as:

$$\text{CSI} = \frac{\sum_{i=1}^{n_e} r_{e_i} w_{t_i}}{\sum_{i=1}^{n_e} w_{t_i} + \lambda(n_e^- + \alpha n_{\text{gap}})} \quad (1)$$

where n_e^- is the number of neutral or transactional messages in the conversation, n_{gap} is the number of long communication gaps, and (α, λ) are non-negative hyperparameters that control the strength of the penalties associated with these phenomena. The numerator in (1) aggregates emotional evidence across the conversation in a time-weighted fashion. Each r_{e_i} summarizes the sentiment of message i on a continuous scale from -1 (strongly negative) to $+1$ (strongly positive), while w_{t_i} encodes how recent that message is at the time CSI is queried. We adopt an exponential decay function,

$$w_{t_i} = \exp(-\beta \Delta t_i), \quad \Delta t_i = T_{\text{now}} - t_i, \quad (2)$$

where t_i is the timestamp of message i (in seconds), T_{now} is the query time, and $\beta > 0$ is a decay rate. When $\Delta t_i = 0$, the weight is $w_{t_i} = 1$ and the message contributes fully to the index; as Δt_i grows, the influence of that message vanishes smoothly. The choice of an exponential kernel enables an efficient $O(1)$ aggregation scheme, as detailed in Section IV.

The denominator of (1) plays a dual role. The first term, $\sum_i w_{t_i}$, normalizes the weighted emotional sum. The second term,

$$\lambda(n_e^- + \alpha n_{\text{gap}}), \quad (3)$$

penalizes patterns associated with disengagement: n_e^- counts neutral messages ($|r_{e_i}| \leq \tau_{\text{neutral}}$), while n_{gap} counts long communication gaps (intervals exceeding T_{gap} , e.g., 24 hours). The hyperparameter λ scales the penalty impact, while α controls the relative severity of gaps versus neutral messages. This structure ensures that CSI reflects both explicit sentiment and silent dissatisfaction signals.

The CSI scores are mapped into five satisfaction levels: *Very Satisfied* $[0.7, 1.0]$, *Satisfied* $[0.3, 0.7]$, *Neutral* $[-0.3, 0.3]$, *Dissatisfied* $[-0.7, -0.3]$, and *Very Dissatisfied* $[-1.0, -0.7]$. This empirical framework translates numerical scores into operational insights, allowing for domain-specific threshold adjustments based on service requirements.

B. Message-Level Sentiment and Neutrality

The emotional intensity r_{e_i} in the Equation (1) is derived by mapping customer messages to scalar values. Messages are processed via a pipeline combining heuristics, supervised classifiers, or LLMs to assign sentiment labels. Each label j carries a calibrated weight $w_j \in [-1, 1]$, where values near $+1$ represent strong positive affect, -1 signify strong negative affect, and 0 denote ambiguous or neutral sentiment. This weighting ensures r_{e_i} accurately reflects both polarity and emotional strength. Given the set of labels attached to message i , with cardinality k , we define the emotional intensity as the simple average:

$$r_{e_i} = \begin{cases} \frac{1}{k} \sum_{j=1}^k w_j, & k > 0, \\ 0, & k = 0. \end{cases} \quad (4)$$

This formulation projects multi-label messages onto a continuous scale, with unconfident assignments defaulting to neutral. To isolate substantive content from routine chatter, we apply a neutrality threshold $\tau_{\text{neutral}} \in [0, 1)$. Messages satisfying $|r_{e_i}| \leq \tau_{\text{neutral}}$ are treated as neutral, contributing to the penalty term n_e^- in equation (3) rather than the emotional numerator of (1). Conversely, messages exceeding this threshold are integrated into the weighted affective sum $\sum r_{e_i} w_{t_i}$. Since messages typically contain few labels, this mapping remains computationally efficient.

In addition to neutrality, we explicitly track communication gaps at the message level. Let t_{i-1} and t_i denote the timestamps of two consecutive customer messages in the same conversation. If the inter-arrival time

$$\Delta t_{\text{gap}} = t_i - t_{i-1} \quad (5)$$

When a message delay exceeds a threshold T_{gap} (e.g., 24 hours), the gap counter n_{gap} increments. This mechanism captures prolonged silence which, unlike isolated neutral messages, often signals disengagement or unresolved dissatisfaction—especially following negative or ambiguous exchanges.

Figure 1 illustrates these dynamics: historical sentiment (Msg 1) decays over time, while communication gaps (Msg 2–3) are explicitly flagged. By amplifying the denominator in (3), such gaps and neutral clutter lower the CSI even in the absence of negative text. Thus, our framework ensures that explicit sentiment drives the index directly via (4), while silent disengagement is captured indirectly through structured penalties.

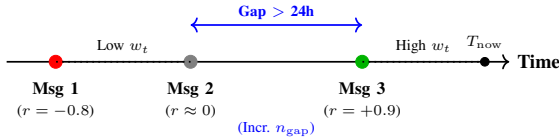


Fig. 1. Timeline illustrating emotional polarity and communication gaps.

C. Hyperparameter Selection and Calibration

CSI (1) is governed by three hyperparameters: α (gap weighting), β (temporal decay), and λ (penalty sensitivity), which balance recency, communication gaps, and neutral content. The decay rate β in $w_{t_i} = e^{-\beta \cdot \Delta t_i}$ dictates how quickly influence diminishes; larger β values increase responsiveness to recent interactions. Calibrating these parameters ensures that theoretical objectives align with empirical domain requirements. To select an appropriate β , we consider typical conversation dynamics in social media customer service, where interactions often span days or weeks. Based on preliminary analysis of sentiment shifts, we set $\beta = 1.0 \times 10^{-6} \text{ s}^{-1}$ for practical implementation. This choice

implies that a message retains approximately 50% of its original weight after a period (half-life) of roughly 8 days:

$$T_{1/2} = \frac{\ln(2)}{\beta} = \frac{0.693}{1.0 \times 10^{-6}} \approx 693,147 \text{ s} \approx 8.02 \text{ days}$$

This parameter choice ensures that messages from the past week retain substantial influence, whereas messages older than one month decay to negligible weights ($e^{-2.6} \approx 0.07$), effectively removing outdated context from the current CSI. The parameter α scales communication gaps relative to neutral messages in the penalty $\lambda(n_e^- + \alpha n_{\text{gap}})$. We set $\alpha = 1.0$ to treat gaps and neutral messages equally, ensuring a conservative weighting that avoids biasing against benign asynchronous habits. The sensitivity parameter λ controls the penalty's impact; at $\lambda = 0$, disengagement is ignored. Given that 30–40% of our dataset consists of neutral messages, sensitivity analysis identified $\lambda = 0.03$ as optimal. This trade-off accounts for 10–20% of the total score, penalizing sparse patterns without overwhelming the core emotional signals.

Based on our experimental analysis, we established a set of robust parameters, including $\beta = 10^{-6} \text{ s}^{-1}$, $\alpha = 1.0$ (gap multiplier), $\lambda = 0.03$, $\tau_{\text{neutral}} = 0.05$, and $T_{\text{gap}} = 86400 \text{ s}$ (24-hour gap threshold). These values demonstrated the best fit for capturing customer sentiment dynamics in our test scenarios. To prevent overfitting on the modest pilot sample, these hyperparameters were selected via coarse grid search and qualitative inspection. We confirmed that performance remains stable on a flat plateau around these values (Section IV). Rather than universal constants, we view (α, λ, β) as deployment knobs to be recalibrated for different organizational contexts or service norms.

D. Survey Instrument and Overall Satisfaction

To establish a ground truth for validation, we utilize a standardized survey combining three constructs: (i) **CSAT** (1–5 Likert) for immediate service evaluation, (ii) **NPS** (0–10) for long-term loyalty [17], and (iii) a **Happiness Index** (1–10) to capture affective trust [18]. All scores are normalized to a 1–10 scale ($\text{CSAT}_{\text{norm}}$, NPS_{norm} , Happiness). The composite **Overall Satisfaction** score is defined by an equal-weighted average:

$$\text{Overall} = \frac{1}{3}(\text{CSAT}_{\text{norm}} + \text{NPS}_{\text{norm}} + \text{Happiness}) \quad (6)$$

This aggregate serves as our primary survey-based metric, ensuring a balanced representation of cognitive and affective satisfaction without privileging any single construct.

IV. PILOT EXPERIMENTAL RESULTS

We evaluate CSI on real-world chat logs from 100 eligible customers (inclusion criteria: ≥ 10 messages, 2–3 weeks activity), resulting in 36 fully completed CSI–survey pairs (36% completion rate). For validation, a standardized survey combined CSAT (1–5), NPS (0–10), and a Happiness Index into a composite *Overall Satisfaction* score (Equation 6), normalized to a 1–10 scale to serve as the ground-truth benchmark. CSI

TABLE I
DESCRIPTIVE STATISTICS FOR CSI AND OVERALL SATISFACTION SCORES

Statistic	CSI	Overall Satisfaction
Mean	6.80	6.80
SD	1.85	2.89
Median	7.26	7.38
Range	[2.56, 8.94]	[1.63, 10.63]
IQR	2.45	4.25

is maintained online via exponential aggregation by storing running totals: $BaseScoreSum$ ($\sum r_{e_i} e^{\beta t_i}$), $BaseWeightSum$ ($\sum e^{\beta t_i}$), $NeutralCount$ (n_e^-), and $GapCount$ (n_{gap}). Each new message triggers an $O(1)$ update to these counters. Upon query, the system applies the decay factor $e^{-\beta T_{now}}$ and the penalty $\lambda(n_e^- + \alpha n_{gap})$ to compute the final CSI in $O(1)$ time. This architecture enables real-time monitoring in high-volume environments without rescanning message history.

A. Evaluation Metrics

We assess how closely CSI aligns with survey-based satisfaction using four views: (i) Pearson and Spearman correlations against Overall Satisfaction (target $r > 0.6$); (ii) linear regression $Survey = \beta_0 + \beta_1 \cdot CSI$ with R^2 , RMSE, and MAE; (iii) classification of satisfaction tiers (High/Medium/Low) with accuracy, F1, and ROC/AUC for a binary satisfied vs. not-satisfied split. Robustness is examined by stratifying results across channels and conversation volumes (High ≥ 50 messages, Medium 20–49, Low ≤ 20).

In addition we probe sensitivity to the penalty and decay hyperparameters by varying $\alpha \in \{0.5, 1.0, 2.0\}$, $\lambda \in \{0.01, 0.03, 0.05\}$, and $\beta \in \{0.5, 1.0, 2.0\} \times 10^{-6}$. For each setting we report changes in correlation (Δr), R^2 , and AUC. A no-penalty variant ($\lambda = 0$) is contrasted with the full model to quantify the contribution of silent-dissatisfaction terms, and channel-specific slices reveal whether fast vs. slow channels react differently to β and T_{gap} .

B. Distributional Characteristics and Relationships

Figure 2 and Table I summarize the distributions of CSI and Overall Satisfaction. The CSI (Figure 2a) shows a mean of 6.80 (SD = 1.85, range: 2.56–8.94) with a right-skewed pattern, effectively capturing positive service experiences across the spectrum. In contrast, the Overall Satisfaction (Figure 2b) exhibits the same mean (6.80) but higher variability and a bimodal distribution peaking at the extremes (2–3 and 9–10).

This bimodality aligns with theoretical expectations that customers often hold polarized rather than neutral reflective positions. While the identical means suggest significant alignment, the distributional differences highlight their complementary nature: CSI reflects continuous conversational dynamics, whereas surveys capture discrete, polarized assessments. The relationship between CSI and Overall Satisfaction was examined using both parametric and non-parametric correlation measures. Pearson correlation analysis revealed a strong positive linear relationship ($r =$

0.817, $p < 0.001$), indicating that 66.7% of the variance in Overall Satisfaction can be explained by CSI scores. This correlation strength ($r > 0.8$) exceeds conventional thresholds for strong associations in behavioral research and demonstrates.

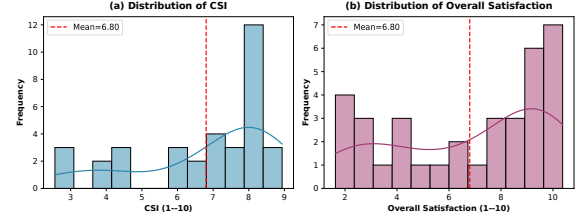


Fig. 2. Score distributions: (a) CSI (mean=6.80, SD=1.85) showing a right-skewed pattern; (b) Overall Satisfaction (mean=6.80, SD=2.89) exhibiting bimodality. Identical means confirm strong central alignment between metrics.

Spearman rank correlation ($\rho = 0.772, p < 0.001$) confirms a robust, linear relationship between CSI and satisfaction, unaffected by outliers. As shown in Figure 3, CSI correlates with all survey components: Happiness ($r = 0.862$), NPS ($r = 0.792$), and CSAT ($r = 0.766$). The particularly high correlation with Happiness suggests that CSI effectively captures the emotional and experiential dimensions of service, validating its design focus on sentiment intensity and temporal dynamics. The inter-correlations among survey components (all $r \geq 0.921$) demonstrate high internal consistency of the survey instrument, while CSI’s moderate-to-strong correlations with each component indicate that it captures distinct but related aspects of customer satisfaction.

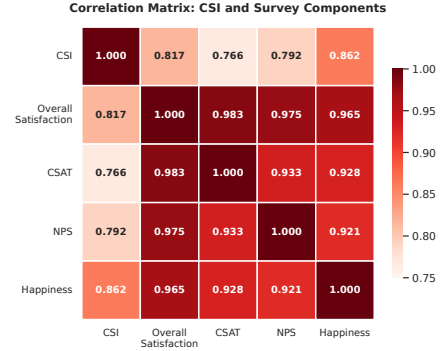


Fig. 3. Correlation matrix between CSI and survey components. CSI shows strong correlations with Happiness ($r = 0.862$), NPS ($r = 0.792$), and CSAT ($r = 0.766$), indicating effectiveness in capturing emotional and experiential dimensions of satisfaction.

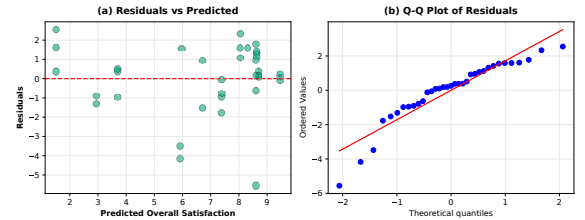


Fig. 4. Regression diagnostics: (a) Residual plot showing random scatter, confirming linearity ($\pm RMSE$ shown); (b) Q-Q plot indicating approximate normality. These validate the linear model and statistical inferences.

Additionally, linear regression analysis was performed to quantify the predictive relationship between CSI and Overall Satisfaction. The fitted model (Figure 5) is:

$$\text{Overall Satisfaction} = -1.661 + 1.243 \times \text{CSI} \quad (7)$$

The model demonstrates excellent fit ($R^2 = 0.667$, $F(1, 34) = 65.4$, $p < 0.001$), with CSI accounting for 66.7% of the variance in Overall Satisfaction. The significant positive coefficient ($\beta = 1.243$, $t = 8.09$, $p < 0.001$) confirms that CSI is a strong predictor of survey outcomes. Residual diagnostics (Figure 4) validate the regression assumptions: the residual plot shows no systematic patterns, confirming linearity, while the Q-Q plot indicates approximate normality with only minor tail deviations. The root mean squared error (RMSE = 1.734) and mean absolute error (MAE = 1.272) provide practical benchmarks for prediction accuracy. These error metrics, relative to the scale range (1–10), indicate that CSI predictions are within approximately 1.3–1.7 points of actual satisfaction scores on average, which is acceptable for practical applications in customer service management.

C. Synthesis and Implications

The comprehensive validation analysis demonstrates that the proposed CSI framework achieves strong empirical alignment with traditional survey-based satisfaction measures across multiple evaluation dimensions. Pearson correlation analysis revealed a robust linear relationship ($r = 0.817$), with the linear regression model explaining a substantial proportion of the variance in Overall Satisfaction ($R^2 = 0.667$, RMSE = 1.734). The practical implications are significant: customer service organizations can leverage CSI for real-time satisfaction monitoring without survey fatigue, proactive identification of at-risk customers through automated classification, continuous measurement across all customer interactions rather than periodic sampling, and integration with automated service recovery systems for immediate intervention. Together, these results establish CSI as a validated, computationally efficient framework for real-time customer satisfaction assessment in multi-channel service environments.

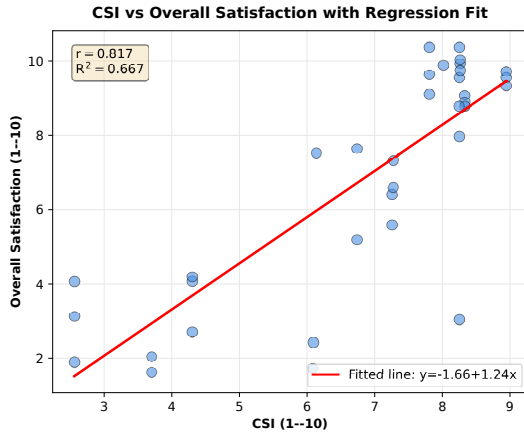


Fig. 5. Linear relationship between CSI and Overall Satisfaction. Strong positive correlation ($r = 0.817$, $p < 0.001$) with $R^2 = 0.667$.

V. DISCUSSION

Beyond the aggregate performance reported above, ablation analyses provide additional insight into the internal structure of the proposed CSI. As summarized in Table II, removing the penalty term entirely ($\lambda = 0$) substantially degrades performance, with Pearson correlation dropping from $r = 0.817$ to $r = 0.758$ and R^2 decreasing by 13.7% (from 0.667 to 0.576). This reduction demonstrates that silent-dissatisfaction signals—captured through neutral-message counts and communication gaps—contribute meaningfully to satisfaction prediction beyond raw sentiment polarity alone. In other words, CSI is not merely a re-scaled sentiment average, but a composite index in which disengagement signals play a measurable role in aligning the index with survey-based satisfaction.

Sensitivity experiments reveal that CSI is robust to moderate misspecification of penalty parameters. Varying λ (0.01–0.05) and α (0.5–2.0) maintains strong performance ($r > 0.80$, $R^2 > 0.65$, AUC > 0.93), with the baseline ($\lambda = 0.03$, $\alpha = 1.0$) being optimal. This flat performance surface allows practitioners to calibrate penalties to specific service norms without jeopardizing validity, though excessive penalties slightly degrade sensitivity to positive sentiment. In contrast, CSI is more sensitive to the decay rate β . Both halving ($r = 0.781$, $R^2 = 0.611$) and doubling ($r = 0.769$, $R^2 = 0.593$) underperform the baseline. These patterns align with psychological recency effects: recent interactions must be prioritized without overshadowing the broader history. Extreme β values either collapse the index into a simple average or over-emphasize final messages, deteriorating model fit. In our chat setting, a half-life of 8 days proved optimal, though longer service cycles (e.g., banking) may warrant slower decay. Overall, these findings establish CSI as a validated, efficient framework. The robustness to penalty variations offers operational flexibility, while the sensitivity to β highlights the necessity of aligning temporal weighting with industry-specific interaction patterns.

As summarized in Table III, our CSI compares favorably with prior sentiment-based satisfaction indices. Studies correlating Net Sentiment Score or social media polarity with NPS/CSAT typically report moderate associations ($r \approx 0.4$ – 0.7 , $R^2 \approx 0.25$ – 0.45) [19]. Aspect-based approaches for review rating prediction achieve stronger fits ($R^2 \approx 0.40$ – 0.55) and 70–85% classification accuracy [20], [21], while sentiment classifiers in service domains attain AUC values of 0.85–0.90. In contrast, our time-weighted, penalty-augmented CSI achieves a higher correlation ($r = 0.817$, $R^2 = 0.667$) and an AUC of 0.943, placing it at the upper end of literature-reported performance. These results prove that modeling conversational recency and silent dissatisfaction aligns more tightly with self-reported happiness than traditional, static metrics. By evaluating multi-turn conversations rather than independent posts, CSI successfully integrates temporal decay and penalties for neutrality or communication gaps, offering a more faithful and operationally useful proxy for satisfaction.

TABLE II
ABLATION AND HYPERPARAMETER SENSITIVITY OF CSI

Variant	Pearson r	R^2	AUC
Proposed model (baseline)	0.817	0.667	0.943
Low penalty ($\lambda = 0.01, \alpha = 0.5$)	0.815	0.665	0.941
High λ , low α ($\lambda = 0.05, \alpha = 0.5$)	0.810	0.658	0.937
High penalty ($\lambda = 0.05, \alpha = 2.0$)	0.805	0.656	0.937
No penalty ($\lambda = 0$)	0.758	0.576	0.915
Low decay ($\beta/2$)	0.781	0.611	0.927
High decay (2β)	0.769	0.593	0.918

TABLE III
CSI VS. PRIOR SENTIMENT-BASED INDICES

Study	Performance
Ours	$r = 0.817$; $R^2 = 0.667$; AUC = 0.943; mean bias = -0.152
NSS vs. NPS [18]	Moderate correlation ($r = 0.443$)
ABSA-based [19, 22]	$R^2 \approx 0.40$ – 0.55 ; accuracy ≈ 70 – 85%
Digital banking [3]	AUC ≈ 0.85 – 0.90 ; $F1 \approx 0.80$ – 0.88

VI. CONCLUSION

This paper validated a conversation-level, time-weighted Customer Satisfaction Index (CSI) integrating recency, sentiment intensity, and silent dissatisfaction (neutral burden, long gaps) via an $O(1)$ recursive exponential aggregation scheme. Tested on 36 customer-survey pairs, the framework aligns strongly with self-reported satisfaction ($r = 0.817$, $R^2 = 0.667$) with high discriminative power (AUC = 0.943), proving that temporal dynamics and silence yield a superior proxy over static sentiment aggregates. However, generalizability is bounded by a pilot-scale sample ($n = 36$), a single-organization context, and reliance on one multilingual sentiment pipeline. Future work will expand evaluations across diverse domains, calibrate decay parameters by channel, integrate advanced ABSA/LLM architectures, and deploy online A/B tests to measure causal impacts. Additionally, incorporating multimodal signals (voice tone, emojis, attachments) and adaptive decay horizons represents a promising avenue to enhance system robustness.

REFERENCES

- [1] F. V. Morgeson, G. T. M. Hult, U. Sharma, and C. Fornell, "The american customer satisfaction index (acsi): A sample dataset and description," *Data in Brief*, vol. 48, p. 109123, Jun. 2023. [Online]. Available: <http://dx.doi.org/10.1016/j.dib.2023.109123>
- [2] N. Raassens and H. Haans, "Nps and online wom: Investigating the relationship between customers' promoter scores and ewom behavior," *Journal of Service Research*, vol. 20, no. 3, p. 322–334, Mar. 2017. [Online]. Available: <http://dx.doi.org/10.1177/1094670517696965>
- [3] F. V. Morgeson, G. T. M. Hult, U. Sharma, and C. Fornell, "The american customer satisfaction index (acsi): A sample dataset and description," *Data in Brief*, vol. 48, p. 109123, Jun. 2023. [Online]. Available: <http://dx.doi.org/10.1016/j.dib.2023.109123>
- [4] W. Zhang, X. Li, Y. Deng, L. Bing, and W. Lam, "A survey on aspect-based sentiment analysis: Tasks, methods, and challenges," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 11, p. 11019–11038, Nov. 2023. [Online]. Available: <http://dx.doi.org/10.1109/TKDE.2022.3230975>
- [5] Y. C. Hua, P. Denny, J. Wicker, and K. Taskova, "A systematic review of aspect-based sentiment analysis: domains, methods, and trends," *Artificial Intelligence Review*, vol. 57, no. 11, Sep. 2024. [Online]. Available: <http://dx.doi.org/10.1007/s10462-024-10906-z>
- [6] L. Ashbaugh and Y. Zhang, "A comparative study of sentiment analysis on customer reviews using machine learning and deep learning," *Computers*, vol. 13, no. 12, p. 340, Dec. 2024. [Online]. Available: <http://dx.doi.org/10.3390/computers13120340>
- [7] S. Shuqair, R. Filieri, G. Viglia, A. S. Mattila, and D. Costa Pinto, "Leveraging online selling through social media influencers," *Journal of Business Research*, vol. 171, p. 114391, Jan. 2024. [Online]. Available: <http://dx.doi.org/10.1016/j.jbusres.2023.114391>
- [8] G. Kaur and A. Sharma, "Has: Hybrid analysis of sentiments for the perspective of customer review summarization," *Journal of Ambient Intelligence and Humanized Computing*, vol. 14, no. 9, p. 11971–11984, Feb. 2022. [Online]. Available: <http://dx.doi.org/10.1007/s12652-022-03748-6>
- [9] J. Budhwar and S. Singh, *Sentiment Analysis of Customers Review Using Hybrid Approach*. Springer International Publishing, 2022, p. 671–685. [Online]. Available: http://dx.doi.org/10.1007/978-3-030-95502-1_51
- [10] G. Agag, B. Ali Durrani, Z. Hassan Abdelmoety, M. Mostafa Daher, and R. Eid, "Understanding the link between net promoter score and e-wom behaviour on social media: The role of national culture," *Journal of Business Research*, vol. 170, p. 114303, Jan. 2024. [Online]. Available: <http://dx.doi.org/10.1016/j.jbusres.2023.114303>
- [11] A. Baquero, "Net promoter score (nps) and customer satisfaction: Relationship and efficient management," *Sustainability*, vol. 14, no. 4, p. 2011, Feb. 2022. [Online]. Available: <http://dx.doi.org/10.3390/su14042011>
- [12] T. Ho and V.-H. Nguyen, *Customer Analytics Using Sentiment Analysis and Net Promoter Score*. IGI Global Scientific Publishing, Jan. 2023, p. 1076–1091. [Online]. Available: <http://dx.doi.org/10.4018/978-1-7998-9220-5.ch062>
- [13] H. Lu, H. Tan, C. Li, and X. Xu, *How Restaurant Attributes Affect Customer Satisfaction: A Study Based on Sentiment Analysis, Neural Network Modelling and Kano Model Classification*. Springer Nature Switzerland, 2023, p. 233–241. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-32299-0_20
- [14] P. Xia, W. Jiang, J. Wu, S. Xiao, and G. Wang, "Exploiting temporal dynamics in product reviews for dynamic sentiment prediction at the aspect level," *ACM Transactions on Knowledge Discovery from Data*, vol. 15, no. 4, p. 1–29, Apr. 2021. [Online]. Available: <http://dx.doi.org/10.1145/3441451>
- [15] N.-B.-V. Le and J.-H. Huh, "Applying sentiment product reviews and visualization for bi systems in vietnamese e-commerce website: Focusing on vietnamese context," *Electronics*, vol. 10, no. 20, p. 2481, Oct. 2021. [Online]. Available: <http://dx.doi.org/10.3390/electronics10202481>
- [16] S. Park, J. Cho, K. Park, and H. Shin, "Customer sentiment analysis with more sensibility," *Engineering Applications of Artificial Intelligence*, vol. 104, p. 104356, Sep. 2021. [Online]. Available: <http://dx.doi.org/10.1016/j.engappai.2021.104356>
- [17] F. F. Reichheld, "The one number you need to grow," *Harvard business review*, vol. 81, no. 12, pp. 46–54, 124, 2003. [Online]. Available: <https://api.semanticscholar.org/CorpusID:1681767>
- [18] A. Alqurafi and T. Alsanoosy, "Measuring customers' satisfaction using sentiment analysis: Model and tool," *Journal of Computer Science*, vol. 20, no. 4, p. 419–430, Apr. 2024. [Online]. Available: <http://dx.doi.org/10.3844/jcssp.2024.419.430>
- [19] S. T. M. van Velthoven, "Sentiment analysis on social media to predict net promoter score," Master's thesis, Eindhoven University of Technology, Eindhoven, The Netherlands, oct 2014, supervisors: Dr.ir. R.M. Dijkman and Dr. A.M. Wilbik; Program: Master of Science in Operations Management and Logistics; Award date: 2014.
- [20] Y. Wang, Y. Huang, and M. Wang, *Aspect-Based Rating Prediction on Reviews Using Sentiment Strength Analysis*. Springer International Publishing, 2017, p. 439–447. [Online]. Available: http://dx.doi.org/10.1007/978-3-319-60045-1_45
- [21] L. Zhu, M. Xu, Y. Bao, Y. Xu, and X. Kong, "Deep learning for aspect-based sentiment analysis: a review," *PeerJ Computer Science*, vol. 8, p. e1044, Jul. 2022. [Online]. Available: <http://dx.doi.org/10.7717/peerj-cs.1044>